

NAMIT SINGH SARNA

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Final-year Computer Science student | AI, RAG and full-stack builder | Right to work in the UK from August 2026 via Graduate Route

PROFILE

Final-year BSc Computer Science student at the University of Leicester, predicted First Class, with practical experience building AI/RAG systems, full-stack web platforms and cloud-backed applications. I enjoy taking ideas from early concept to working product, including a production Next.js property platform delivered solo for a paying client. I am particularly interested in practical software systems that are usable, maintainable and grounded in real user needs.

CORE SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL

AI / Retrieval: RAG, ChromaDB, OpenAI API, Ollama, embeddings, LLM orchestration

Backend: FastAPI, Spring Boot, REST APIs, MySQL, Redis, WebSockets

Frontend: Next.js, React, Tailwind CSS, Streamlit

Cloud / Tools: Docker, Kubernetes/GKE, Git/GitLab, Linux, pytest

Working style: Agile/Scrum, documentation, problem-solving

EXPERIENCE

Cloud Seven Realty - Freelance Web Developer | Solo client project, 2025-2026

Stack: Next.js, TypeScript, Tailwind CSS

- Designed, built and launched a production real estate website end to end for a paying client, taking ownership from planning through delivery.
- Created a spreadsheet-driven listings workflow so non-technical staff can update property content without developer involvement.
- Optimised the site for mobile usability, page speed, maintainability and long-term client-side upkeep; handed over clear documentation.

AI Education and Digital Transformation Micro-Internship | University of Leicester, July 2025

- Led technical work in a cross-disciplinary team exploring responsible generative AI use in higher education.
- Built learning-assistant prototypes and supported non-CS teammates with engineering decisions and technical delivery.
- Contributed recommendations on AI adoption, academic integrity and student-centred digital transformation.

Student Representative | University of Leicester, May 2025

- Selected to represent Indian students during the official visit of Dr Venkatachalam Murugan, Consul General of India to the UK.
- Participated in roundtable discussion with university leadership on international student experience and collaboration.

FEATURED PROJECTS

UniWise - Retrieval-Augmented Academic Assistant | Final Year Project, 2025-2026

Stack: Python, ChromaDB, OpenAI API, Ollama, Streamlit, pytest

- Built a multi-backend RAG system supporting both OpenAI-hosted models and local Ollama models through a pluggable adapter/factory layer.
- Implemented two-stage retrieval using bi-encoder dense search over ChromaDB with cross-encoder reranking to improve answer grounding.
- Designed a confidence-based refusal mechanism, achieving 1.0 precision, recall, and F1 for out-of-scope rejection across all four evaluated configurations on a 30-question benchmark.
- Wrote 52 pytest unit tests across 10 modules, with all tests passing and 84% overall coverage, including 100% coverage on core logic.
- Built a Streamlit interface with three pedagogical response modes and a telemetry dashboard using K-Means query clustering.

Distributed Whiteboard on Private Cloud | CO3219 module project, Dec 2025

Stack: Docker, Kubernetes/GKE, Redis pub/sub, WebSockets

- Built a real-time collaborative whiteboard using WebSocket synchronisation and Redis pub/sub for multi-instance communication.
- Deployed to a private Kubernetes cluster with load balancing, horizontal scaling and graceful recovery after pod restart.

Spring Boot & Spring Data Backend Project | Jan 2025

Stack: Java, Spring Boot, Spring Data, Gradle, Junit

- Developed a backend web application with RESTful API design and MVC architecture.
- Implemented database operations using Spring Data for cleaner CRUD functionality.
- Achieved 95% module grade.

EDUCATION

BSc Computer Science | University of Leicester, 2023-July 2026

- Predicted First Class Honours; achieved First Class in Year 2.
- Relevant modules include Big Data and Predictive Analytics, Software Architecture, Algorithms and Data Structures, Database Systems, Web and Mobile Development, AI and Data Mining, and Internet and Cloud Computing.

ADDITIONAL PROJECTS

Course Companion Web App | Spring Boot, MySQL, 2025

- Built login, dashboard, XP ring and streak tracker features; contributed around 40% of the codebase across two Agile sprints with 40+ GitLab commits.
- Designed global CSS for 15+ pages, improving responsive layout and UI consistency across the project.

Crime Data Analysis and Prediction | Python, scikit-learn, 2026

- Analysed 1M+ UK Metropolitan Police records using pandas and NumPy, including cleaning, aggregation and feature engineering.

- Trained and evaluated Linear Regression, Decision Tree and Random Forest models, and used K-Means clustering to explore spatial crime patterns.

ACHIEVEMENTS AND LANGUAGES

- Winners, International Student Consultancy Challenge 2025; co-led the Global Career Navigator pitch for international students.
- Meritorious Team Award, CMS Mathematical Modelling Competition 2025; led a Newsvendor inventory model that supported pricing and output recommendations.
- [Contributor to guidance on responsible Generative AI use in learning, with focus on academic integrity, critical thinking and ethical adoption.](#)
- PWN2PLAY Silicon Siege CTF, De Montfort University; contributed across OSINT, cipher analysis, LFSR and geolocation challenges.
- Languages: English fluent; Hindi, Urdu, Punjabi and Kashmiri; German beginner.